

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY
COMPUTER ARCHITECTURE
MODEL QUESTION PAPER

Time : 3 hours

(Maximum marks : 100)

Marks

PART – A

(Maximum marks : 10)

I Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define word.
2. Define cylinder.
3. List the contents of Program Status Word.
4. List the two basic tasks performed by microprogrammed control unit.
5. Define stripping.

(5 x 2 = 10)

PART – B

(Maximum marks : 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Explain DDR SDRAM.
2. Explain the control and timing function of the I/O module.
3. Describe about indirect cycle.
4. Explain parallel processing.
5. Write short note on cache memory principles.
6. Write short note on CD-RW.
7. Describe execute cycle.

(5 x 6 = 30)

PART – C

(Maximum marks : 60)

(Answer one full question from each module. Each question carries 15 marks)

MODULE – I

- III a. Explain I/O function of computer.
b. Explain direct mapping.

7

8

OR

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| IV | a. Explain bus structure. | 8 |
| | b. Explain Rambus DRAM. | 7 |

MODULE – II

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|---|---|---|
| V | a. Compare CAV and multi zoned recording. | 8 |
| | b. Write short note on external devices | 7 |

OR

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| VI | a. Explain CD operation with diagram. | 8 |
| | b. Explain programmed I/O. | 7 |

MODULE – III

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| VII | a. Explain user-visible registers. | 8 |
| | b. Explain indirect cycle. | 7 |

OR

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| VIII | a. Explain control and status registers. | 8 |
| | b. Explain instruction pipelining. | 7 |

MODULE – IV

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|----|--------------------------------------|---|
| IX | a. Explain hardwired implementation. | 8 |
| | b. Explain parallel processing. | 7 |

OR

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| X | a. Explain micro instruction format. | 7 |
| | b. Explain parallel processor organization. | 8 |